ICSA 2026 Short Course

Lecture 3: Covariate shift and density-ratio reweighting

Statistical Foundations of Transfer Learning

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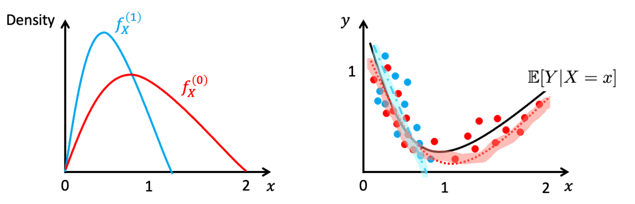
Overview

§3.1: Covariate shift

Covariate shift, formally

In L2 the covariate-shift gap was a fixed penalty $2\,d_{\mathrm{TV}}$ *we could only bound; this lecture shows importance weighting removes it and recovers oracle rates.*

- Source $(X,Y) \sim \mathbb{P}^{(1)}$, target $(X,Y) \sim \mathbb{P}^{(0)}$, on $\mathcal{X} \times \mathcal{Y}$.
- **Covariate shift**:
$$\mathbb{P}_X^{(0)} \neq \mathbb{P}_X^{(1)}, \qquad \mathbb{P}_{Y|X}^{(0)} = \mathbb{P}_{Y|X}^{(1)} \quad (\text{a.s. } \mathbb{P}_X^{(0)}).$$
- Equivalent: there exists a regression / decision function f^* *shared* by source and target; only the feature distribution changes.



Three motivating scenarios

- **Self-driving cars:** training routes $\subsetneq$ deployment routes. Same physics; new lighting, weather, traffic distributions.
- **Medical imaging:** Alzheimer's from MRI: a research-study scanner (source) vs. a community-hospital scanner (target) — same disease, shifted images (our L1 running example).
- **A/B testing for ML:** the production traffic mix shifts after a model is shipped, but the conditional model of $Y \mid X$ is unchanged.

Why special: in all three, the *physics/biology/intent* is fixed — only the *distribution of situations* changes. That is covariate shift.

Density ratio: the central object

Define the source-to-target density ratio

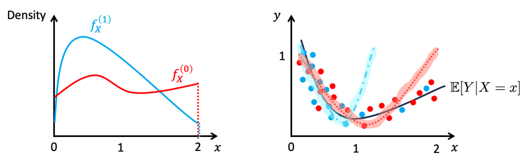
$$w(\boldsymbol{x}) \;=\; \frac{d\mathbb{P}_X^{(0)}}{d\mathbb{P}_X^{(1)}}(\boldsymbol{x}),$$

assuming $\mathbb{P}_X^{(0)} \ll \mathbb{P}_X^{(1)}$ (target absolutely continuous w.r.t. source).

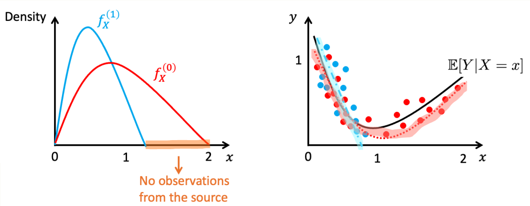
- Intuition: $w(\boldsymbol{x}) > 1$ at points the target "cares about" more than the source.
- Failure case: $\sup_{\boldsymbol{x}} w(\boldsymbol{x}) = \infty$. Then any source-only estimator can be fooled.
- Key fact: for any integrable g, $\mathbb{E}_{\mathbb{P}^{(0)}}[g(X)] = \mathbb{E}_{\mathbb{P}^{(1)}}[w(X)g(X)]$.
- *Think of it as:* reweighting source points by w lets you "pretend" you sampled from the target — importance sampling over $\mathcal{X}$.

Bounded vs. unbounded density ratios

Bounded w (well-behaved)



Unbounded w (failure)



Bounded w + small effective sample size $n_1/\|w\|_\infty \Rightarrow$ usable transfer.

Unbounded w: a single source point in a region the source rarely visits but the target loves gets an astronomically large weight — one observation dominates the empirical risk.

Check-in: what defines covariate shift?

In which of the following scenarios is the assumption $\mathbb{P}^{(0)}_{Y|X} = \mathbb{P}^{(1)}_{Y|X}$ *most plausible?*

(a) Survey of US households 2010 vs. 2020: X is income, Y is voting preference.

(b) Camera sees same kind of objects under new lighting; X is the pixel array, Y is the object class.

(c) Hospital A's MRI vs. Hospital B's CT scan; Y is whether the patient has tumor.

(d) Pre- vs. post-pandemic search queries; Y is whether to click a particular ad.

Answer

(b). Same physical objects under different lighting – the underlying object-given-pixel mapping is unchanged. (a) and (d) involve a shifting world (people change their preferences), and (c) doesn't even share the feature space (X is different).

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§3.2: Is covariate shift adaptable for free?

When ERM survives covariate shift unaided

Consider parametric MLE under a **well-specified** model: $\mathbb{P}^{(k)}_{Y|X} = p_\theta(Y|X)$ with the same θ for $k = 0, 1$.

- The MLE on source converges to θ^* regardless of $\mathbb{P}^{(1)}_X$ (as long as it puts mass on the relevant directions).
- Plug in $\hat{\theta}$ to get $\hat{p}_{Y|X}$; this works on the target by assumption.
- Conclusion: **well-specified parametric MLE is automatically robust to covariate shift**.
- *Why?* MLE maximizes $\sum_i \log p_\theta(Y_i \mid X_i)$ — a conditional loss; the score $\nabla_\theta \log p_\theta(Y \mid X)$ depends only on $P_{Y|X}$, so the source marginal is irrelevant.
- In fact source-only MLE is *minimax-optimal* on the target under a well-specified model — with *no* bounded-density-ratio assumption (Ge et al., 2024).

When ERM fails: mis-specification or non-parametric

Two regimes where naive ERM is biased on the target:

- **Mis-specified model:** the best linear (etc.) approximation is determined by $\mathbb{P}_X^{(k)}$. Different $\mathbb{P}_X^{(k)} \Rightarrow$ different "best" parameter $\Rightarrow$ different prediction.
- **Non-parametric estimators** (kernel, nearest neighbor): convergence rates depend on the local density of training points. Source-skewed densities give source-skewed accuracy.

Takeaway: "free" covariate-shift adaptation is the exception, not the rule. When in doubt, correct.

Beyond parametric models: k-NN under covariate shift

Suppose f^* is β-Hölder and $\mathbb{P}_X^{(1)}$ has support strictly containing $\mathbb{P}_X^{(0)}$. Then k-NN regression has target risk

$$R^{(0)}(\widehat{h}_{kNN}) - R^{(0)}(f^*) \lesssim \mathbb{E}_{X \sim \mathbb{P}^{(0)}}\left[w(X)^\beta\right] \cdot n_1^{-2\beta/(2\beta+d)}$$

where $w(X) = \frac{d\mathbb{P}_X^{(0)}}{d\mathbb{P}_X^{(1)}}(X)$ is the density ratio (recall $w(X) = d\mathbb{P}_X^{(0)}/d\mathbb{P}_X^{(1)}$, the same object reweighting uses) — large exactly where the source is sparse, so the constant $\mathbb{E}[w^\beta]$ (not the rate) is what covariate shift inflates.

Why w^β? A β-Hölder f^* makes a neighbor at distance r contribute bias $\sim r^\beta$; where source is sparse (w large) the k-NN radius is large, so bias $\sim w^\beta$ — hence $\mathbb{E}[w^\beta]$.

- If $\mathbb{P}_X^{(0)}$ concentrates where source is sparse, $\mathbb{E}[w^\beta]$ blows up.
- This is the non-parametric face of covariate shift: the rate is right, the constant explodes.

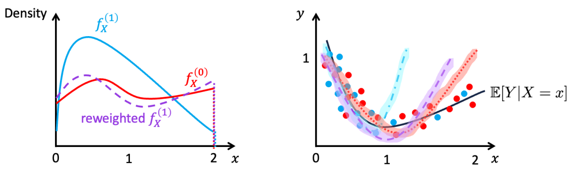
Fix: reweight source points so they mimic the target density — the reweighting method, next.

§3.3: The reweighting method

Reweighted ERM

Fix a loss $\ell(h(X), Y)$ and recall $\mathbb{E}_{\mathbb{P}^{(0)}}[\ell] = \mathbb{E}_{\mathbb{P}^{(1)}}[w(X)\ell]$. Define the **importance-weighted ERM**:

$$\widehat{h}_{\mathrm{IW}} = \arg\min_{h \in \mathcal{H}} \frac{1}{n_1} \sum_{i=1}^{n_1} w(\boldsymbol{x}_i^{(1)})\, \ell\big(h(\boldsymbol{x}_i^{(1)}), y_i^{(1)}\big).$$



Source points reweighted by $w(\boldsymbol{x}_i)$ (larger = "more like target"); after reweighting the empirical law mimics the target.

Why reweighting works

Theorem 3.3.1 (Cortes et al., 2010)

The IW objective is unbiased for the target risk, and the deviation is governed by the second-moment / Rényi divergence $d_2(\mathbb{P}^{(0)} \| \mathbb{P}^{(1)}) = \mathbb{E}_{\mathbb{P}^{(1)}}[w(X)^2]$*: w.p.* $1 - \delta$,

$$R^{(0)}(\widehat{h}_{\mathrm{IW}}) \;\leq\; \min_h R^{(0)}(h) + \mathcal{O}\left(\sqrt{\frac{d_2(\mathbb{P}^{(0)} \| \mathbb{P}^{(1)})\,\mathrm{VC}(\mathcal{H})}{n_1}}\right).$$

- Unbiased: $\mathbb{E}_{\mathbb{P}^{(1)}}[w(X)\ell(h(X), Y)] = R^{(0)}(h)$.
- *Why?* Under covariate shift $\mathbb{P}^{(0)}_{Y|X} = \mathbb{P}^{(1)}_{Y|X}$, so $\mathbb{E}_{\mathbb{P}^{(1)}}[w(X)\ell] = \int w\, d\mathbb{P}^{(1)}_X \int \ell\, d\mathbb{P}^{(1)}_{Y|X} = \int d\mathbb{P}^{(0)}_X \int \ell\, d\mathbb{P}^{(0)}_{Y|X} = R^{(0)}(h)$.

Why reweighting works (cont.)

- "Effective sample size" $= n_1/d_2$: heavy-tailed w (large d_2) wastes data. (Bounded $w \leq W$ is the special case $d_2 \leq W$.)
- *Why d_2?* The IW risk has variance $\propto \mathbb{E}_{\mathbb{P}^{(1)}}[w^2\ell^2]/n_1 \leq \mathbb{E}_{\mathbb{P}^{(1)}}[w^2]/n_1 = d_2/n_1$ — noise scales with the *second moment* of the weight, not its max.
- Unbounded w: the rate degrades below $\sqrt{n_1}$; in practice w is unknown $\Rightarrow$ density-ratio estimation (§3.4).

Check-in: importance weighting

The true density ratio is $w(\boldsymbol{x}) = e^{2\boldsymbol{x}_1}$, *so* $\sup w = \infty$ *on* $\mathbb{R}^d$. *Your source has* $n_1 = 10{,}000$ *points.*

Which fix is principled?

(a) Use IW-ERM; the unbiased estimator ensures consistency.

(b) Clip w at some W, accept a small bias, recover variance control.

(c) Switch to target-only ERM; we have no labels, so we are stuck.

(d) Use both methods and take the minimum-MSE estimator.

Answer

(b). Unbounded w blows up the IW variance: a single source point with very large weight can dominate the empirical risk. Truncating at some level W yields a tractable bias-variance trade-off (also justified theoretically when w has heavy tails).

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Numerical illustration: importance weighting

Numerical illustration: setup

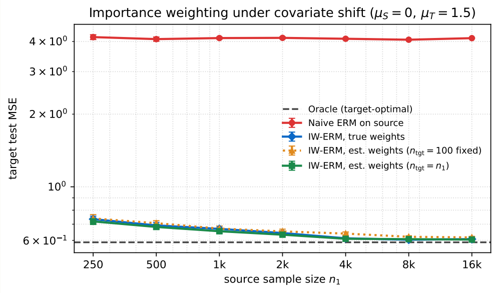
Source $X \sim N(0,1)$, target $X \sim N(1.5,1)$ (so $w(x) = e^{1.5x-1.125}$ [ratio of $N(1.5,1)$ to $N(0,1)$ densities, so $w(x) = \exp(1.5x - 1.5^2/2)$], bounded on bounded sets). Truth $y = 0.5\,x^2 + 0.3\,\varepsilon$, fit class is *linear* (mis-specified).

- Source OLS: $b \approx 0$ (since $\mathrm{Cov}(x, x^2) = 0$ at $\mu_S{=}0$).
- Target OLS: $b \approx 1.5$ (best linear fit around $\mu_T{=}1.5$).

Compare on the target test set against the **oracle** (population target-optimal fit): naive ERM; IW-ERM with *true* weights; and IW-ERM with *estimated* weights (logistic discriminator), using either a small *fixed* unlabeled-target pool ($n_{\mathrm{tgt}}{=}100$, balanced discriminator) or one that *grows* with n_1.

Code: `sims/L3_importance_weighting.py`.

Numerical illustration: result



- Naive ERM is biased by $\sim 4\times$ the irreducible target MSE — more source never helps.
- IW-ERM with the *true* weights recovers the oracle as n_1 grows.
- Estimated weights from a *fixed* target pool (n_{tgt}=100) plateau *above* the oracle — the weights are only as good as that target sample.
- Grow the target with n_1 and that curve drops to the oracle, matching true weights.

Numerical takeaway

- Under covariate shift + mis-specification, naive ERM is *permanently biased*: more source data does not help.
- IW reweighting is the right correction; it converges to oracle at the standard $\sqrt{n_1}$ rate, scaled by the Rényi divergence $d_2(\mathbb{P}^{(0)} \| \mathbb{P}^{(1)}) = \mathbb{E}_{\mathbb{P}^{(1)}}[w^2]$.
- Estimated weights from a binary discriminator work well *given enough unlabeled target* to learn them (previewed in §3.4, the discriminative method).

§3.4: Density-ratio estimation

Estimating the density ratio: four canonical routes

Everything so far assumed w known; now we estimate it from source plus unlabeled target features. The four routes are surrogates for the same w: match moments (KMM), classify domains (discriminative), or fit w directly (KL / least-squares).

Given source $\{\boldsymbol{x}_i^{(1)}\}_{i=1}^{n_1}$ and unlabeled target $\{\boldsymbol{x}_j^{(0)}\}_{j=1}^{n_0}$:

- **Kernel mean matching** (Huang et al., 2006): match RKHS mean embeddings.
- **Discriminative learning** (Bickel et al., 2009): train a classifier of source vs. target; convert probabilities to a ratio.
- **KL-divergence minimization** (Sugiyama et al., 2008): fit a parametric $\hat{w}$ to maximize the KL surrogate.
- **Least-squares** (Kanamori et al., 2009; Nguyen et al., 2010): solve a convex regression for w directly.

(here n_0 counts unlabeled target points — covariate-shift transfer has no target labels; contrast L2/L4 where n_0 was labeled target examples.)

Density-ratio estimation: scope of the four routes (cont.)

The four routes are the canonical, dimension-robust estimators; the rest of this section develops each in turn.

- **Kernel mean matching** — match RKHS mean embeddings (moments).
- **Discriminative** — classify source vs. target, convert to a ratio.
- **KL / least-squares** — fit w directly via a divergence surrogate.

Other approaches (separate KDE, histograms, semi-parametric) are useful in 1-D / low-dim or under structural assumptions; we summarize them in one slide at the end.

Kernel mean matching (KMM)

Idea: choose source weights $\widehat{w}_i$ so that the *reweighted* source mean embedding matches the target mean embedding in an RKHS $\mathcal{H}_K$:

$$\min_{\widehat{\boldsymbol{w}} \geq 0} \left\| \frac{1}{n_1} \sum_{i=1}^{n_1} \widehat{w}_i \, K(\boldsymbol{x}_i^{(1)}, \cdot) - \frac{1}{n_0} \sum_{j=1}^{n_0} K(\boldsymbol{x}_j^{(0)}, \cdot) \right\|_{\mathcal{H}_K}^2 \quad \text{s.t.} \;\; \tfrac{1}{n_1} \sum_i \widehat{w}_i = 1, \; \widehat{w}_i \leq B.$$

- Convex QP in $\widehat{w}$; no parametric model for w.
- Connects directly to MMD (Lecture 2): the objective *is* an empirical squared MMD.
- *Picture:* treat $K(\boldsymbol{x}, \cdot)$ as a "fingerprint" of $\boldsymbol{x}$; KMM finds weights so the weighted-average source fingerprint equals the target's — they look identical to any $f \in \mathcal{H}_K$.

KMM: theory

KMM is an algorithm (Huang et al., 2006); its *consistency* is analyzed later:

Theorem 3.5.1 (Gretton et al., 2009; Yu and Szepesvári, 2012)

With a universal kernel (i.e. $\mathcal{H}_K$ dense in $C(\mathcal{X})$, so matching means forces the full distributions to agree) and bounded weights $\widehat{w} \leq B$, the reweighted feature mean converges ($\widehat{w} \to w$ in MMD as $n_1, n_0 \to \infty$) and the IW-ERM with $\widehat{w}$ is consistent.

- Practical note: tune B on a held-out validation set; popular choices $B \in [10, 100]$.
- Strength: works well in moderate dimension; nonparametric.
- Weakness: doesn't extrapolate beyond the source support; sensitive to B.

Discriminative density-ratio learning

Stack source (label $z = 0$) and target (label $z = 1$) and train any binary classifier $\hat{\eta}(\boldsymbol{x}) = \widehat{\Pr}(z = 1 \mid \boldsymbol{x})$. By Bayes, $\Pr(z{=}k \mid \boldsymbol{x}) \propto p_X^{(k)}(\boldsymbol{x}) \Pr(z{=}k)$, so $\frac{p_X^{(0)}}{p_X^{(1)}} = \frac{\Pr(z=0|\boldsymbol{x})}{\Pr(z=1|\boldsymbol{x})} \cdot \frac{\Pr(z=1)}{\Pr(z=0)}$ (source $z{=}0$, target $z{=}1$). Then by Bayes' rule,

$$\frac{\mathbb{P}_X^{(0)}(\boldsymbol{x})}{\mathbb{P}_X^{(1)}(\boldsymbol{x})} = \frac{\Pr(z = 1 \mid \boldsymbol{x})}{\Pr(z = 0 \mid \boldsymbol{x})} \cdot \frac{\Pr(z = 0)}{\Pr(z = 1)} \approx \frac{\hat{\eta}(\boldsymbol{x})}{1 - \hat{\eta}(\boldsymbol{x})} \cdot \frac{n_1}{n_0}.$$

- Any well-calibrated classifier (logistic, GBM, deep net) gives an estimate.
- **Easy to deploy**; the workhorse method in modern practice.
- Extends to non-tabular data (images, text) because the discriminator can.

This is L2's source-vs-target domain classifier, now repurposed to estimate the weight w rather than just measure distance.

Discriminative density-ratio: theory

Theorem 3.5.2 (Bickel et al., 2009, informal)

Suppose the discriminator class is rich enough and trained with $n_1 + n_0$ joint samples; then with high probability,

$$\left(\mathbb{E}_{\mathbb{P}^{(1)}}\left[(\widehat{w} - w)^2\right]\right)^{1/2} \le C\sqrt{\mathrm{VC}(\mathcal{F})/(n_1 + n_0)} + \textit{approximation error of } \mathcal{F}.$$

- Same rate as plain classification on $z \in \{0, 1\}$ – no curse of dimensionality if the discriminator class is well chosen.
- Couples the IW-ERM analysis to standard classification theory.

KL-divergence minimization (KLIEP)

Parametrize $w_\theta(\boldsymbol{x}) \geq 0$ (e.g. $w_\theta(\boldsymbol{x}) = \exp(\theta^\top \phi(\boldsymbol{x}))$). Maximize

$$\frac{1}{n_0} \sum_j \log w_\theta(\boldsymbol{x}_j^{(0)}) \quad \text{s.t.} \quad \frac{1}{n_1} \sum_i w_\theta(\boldsymbol{x}_i^{(1)}) = 1 \ \big(\text{enforces } \mathbb{E}_{\mathbb{P}^{(1)}}[w] = 1, \text{ i.e. } \hat{p}_0 \text{ is a density}\big).$$

Equivalent to minimizing $\mathrm{KL}(\mathbb{P}_X^{(0)} \,\|\, \hat{w} \cdot \mathbb{P}_X^{(1)})$ over the parameteric class.

Sketch: $\mathrm{KL}(p^{(0)} \| \hat{w}\, p^{(1)}) = \mathbb{E}_{p^{(0)}}[\log p^{(0)}] - \mathbb{E}_{p^{(0)}}[\log \hat{w}] - \mathbb{E}_{p^{(0)}}[\log p^{(1)}]$; only the middle term depends on $\hat{w}$, so minimizing KL = maximizing $\frac{1}{n_0} \sum_j \log w_\theta(\boldsymbol{x}_j^{(0)})$ s.t. normalization.

- **KLIEP** (Sugiyama et al., 2008): closed-form for log-linear w_θ, with cross-entropy-type objective.
- Theory: under standard regularity, $\widehat{w}_\theta$ converges at the parametric rate.
- Caveat: parametric form for w may be restrictive, but works well for moderate d.

Least-squares ratio estimation

Direct convex regression for w itself:

$$\min_{w\in\mathcal{F}} \ \tfrac{1}{2}\,\mathbb{E}_{\mathbb{P}^{(1)}}\left[w(X)^2\right] - \mathbb{E}_{\mathbb{P}^{(0)}}\left[w(X)\right] \ = \ \tfrac{1}{2}\,\mathbb{E}_{\mathbb{P}^{(1)}}\left[(w-w^*)^2\right] + \text{const.}$$

Derivation: $\frac{1}{2}\mathbb{E}_{\mathbb{P}^{(1)}}[w^2] - \mathbb{E}_{\mathbb{P}^{(0)}}[w] = \frac{1}{2}\mathbb{E}_{\mathbb{P}^{(1)}}[w^2] - \mathbb{E}_{\mathbb{P}^{(1)}}[w\,w^*] = \frac{1}{2}\mathbb{E}_{\mathbb{P}^{(1)}}[(w-w^*)^2] - \text{const}$, using $\mathbb{E}_{\mathbb{P}^{(0)}}[g] = \mathbb{E}_{\mathbb{P}^{(1)}}[w^* g]$.

Estimable by sample averages: $\frac{1}{2n_1}\sum_i w(\boldsymbol{x}_i^{(1)})^2 - \frac{1}{n_0}\sum_j w(\boldsymbol{x}_j^{(0)})$.

Theorem 3.5.3 (Kanamori et al., 2009; Nguyen et al., 2010)

For $\mathcal{F}$ a kernel class, $\widehat{w}$ converges to w^* in $L^2(\mathbb{P}^{(1)})$ at the kernel-regression rate.

- Convex; supports kernels and neural-net parameterizations alike.
- Often more robust than KL methods when target has heavier tails than source.

Other approaches (one-slide summary)

Method	**Idea**	**When to use**
Separate KDE	Estimate $\hat{p}_0$, $\hat{p}_1$, take ratio.	Low-d, smooth densities.
Histogram-based	Bin the space; ratio = bin frequency.	Univariate; quick & dirty.
Semi-parametric (Qin ’98)	Tilt: $w_\theta = \exp(\theta^\top \phi)$ via empirical likelihood.	Moderate d + structural prior.

Practical guidance.

- Tabular, moderate d: KMM or KLIEP.
- Image / text / large d: discriminative.
- Heavy-tailed target: least-squares.
- Low-d + smooth densities: separate KDE is fine and exploits structure.

Check-in: which method should you pick?

You have 10,000 source images and 1,000 unlabeled target images at $224 \times 224 \times 3$ *resolution. You need* $\widehat{w}$ *for IW-ERM on a downstream image classifier.*

Which method is the obvious first choice?

(a) Separate KDE on $224^2 \times 3$-dimensional pixel space.

(b) KMM with a Gaussian kernel.

(c) Train a CNN to classify source vs. target; convert logits to a ratio.

(d) Histogram on principal components.

Answer

(c), discriminative density-ratio estimation with a deep net. Curse-of-dimensionality kills (a) and (d); KMM is feasible but tuning B and the Gaussian bandwidth in 150,000-D is impractical. A binary classifier directly inherits the deep-net's representation.

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Lecture summary

- Covariate shift = different $\mathbb{P}_X^{()}$, same $\mathbb{P}_{Y|X}^{()}$.
- "Free" adaptation only for well-specified parametric MLE; otherwise ERM is biased.
- Reweight by $w = d\mathbb{P}_X^{(0)}/d\mathbb{P}_X^{(1)}$ to recover the target objective.
- Bounded w + uniform deviation $\Rightarrow$ standard rates with effective sample size n_1/d_2.
- Density-ratio estimation: KMM, discriminative, KL, least-squares (and friends).

Next: Lecture 4 turns to *posterior drift* – $\mathbb{P}_X^{()}$ is shared but $\mathbb{P}_{Y|X}^{()}$ differs – and asks how to bias an estimator toward the source as a route to "safe transfer."

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